HPN-CR: Heterogeneous Parallel Network for SAR-Optical Data Fusion Cloud Removal

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***Abstract*— Synthetic aperture radar (SAR)-optical data fusion cloud removal is a highly promising cloud removal technology that has attracted considerable attention. In this field, primary researches are based on deep learning, which can be divided into two categories: convolutional neural network (CNN)-based and Transformer-based. In the cases of extensive cloud coverage, CNN-based methods, with their local spatial awareness, effectively capture local structural information in SAR data, preserving clear contours of cloud-removed land covers. However, these methods struggle to capture global land cover information in optical images, often resulting in notable color discrepancies between recovered and cloud-free regions. Conversely, Transformer-based methods, with their global modeling capability and inherent low-pass filtering properties, excel at capturing long-range spatial correlations in optical images, thereby maintaining color consistency across cloud-removal outputs. However, these methods are less effective at capturing the fine structural details in SAR data, which can lead to blurred local contours in the final cloud-removed images. In this context, a novel framework called heterogeneous parallel network for cloud removal (HPN-CR) is proposed to achieve high-quality cloud removal under extensive cloud coverage conditions. HPN-CR employs the proposed heterogeneous encoder with its SAR-optical input approach to effectively extract and fuse the local structural information in cloudy areas from SAR images, with the spectral information about land covers in cloud-free areas from the whole optical images. In particular, it uses a ResNet network with local spatial awareness to extract SAR features. It also uses the proposed Decloudformer, which globally models multiscale spatial correlations, to extract optical features. The output features are fused by the heterogeneous encoder and then reconstructed to cloud-removal images through a pixelshuffle-based decoder. Comprehensive experiments were conducted and the experimental results demonstrated the effectiveness and superiority of the proposed method. The code is available at https://github.com/G-pz/HPN-CR.**

***Index Terms*— Cloud removal, convolutional neural network (CNN), data fusion, optical imagery, synthetic aperture radar (SAR)-optical, Transformer.**

I. Introduction

OPTICAL satellite remote sensing has become significant in various ground observation tasks for decades, such as environmental preservation, scene classification, change detection, and monitoring [1], [2]. However, since the light waves cannot penetrate the cloud layer, the optical images acquired by satellite sensors are inevitably affected by cloud contamination [3]. This results in information loss, hindering image interpretation and application, particularly in scenarios demanding seamless temporal monitoring [4]. Therefore, it is essential to preprocess the optical image data by introducing cloud removal to eliminate the influence of clouds.

The methods for cloud removal can be coarsely divided into two categories based on whether synthetic aperture radar (SAR) images are used or not [5]. The first type of method relies exclusively on optical data, using images or bands from cloud-free regions to fill in missing information in cloud-covered areas. Early cloud removal methods were predicated on the assumption that the cloudy region tends to follow a common spectral response with the cloud-free regions, utilizing linear filtering techniques to remove clouds [5], [6]. Subsequently, methods based on multispectral, multitemporal, or multisource [7] optical imageries were developed. They have proven to be effective with low cloud coverage and for relatively short cloud durations [8]. However, in the cases of persistent cloud cover, the unavailability of reference data renders all of these methods ineffective [8].

To address the above issues, the second type of method, SAR-optical data fusion cloud removal techniques, has emerged and gained increasing attention. These techniques aim to reconstruct cloud-free optical data from a cloudy optical input by leveraging auxiliary SAR imagery. Due to the cloud penetration capacity of SAR, it can acquire surface features even under cloud cover, which provides auxiliary information for the cloud removal task. Deep learning is the mainstream method in this field, and according to different basic operators of the network, it can be divided into two main categories: convolutional neural network (CNN)-based approaches [7], [9], [10], [11] and Transformer-based approaches [5], [12]. Methods based on CNN use a convolutional network as

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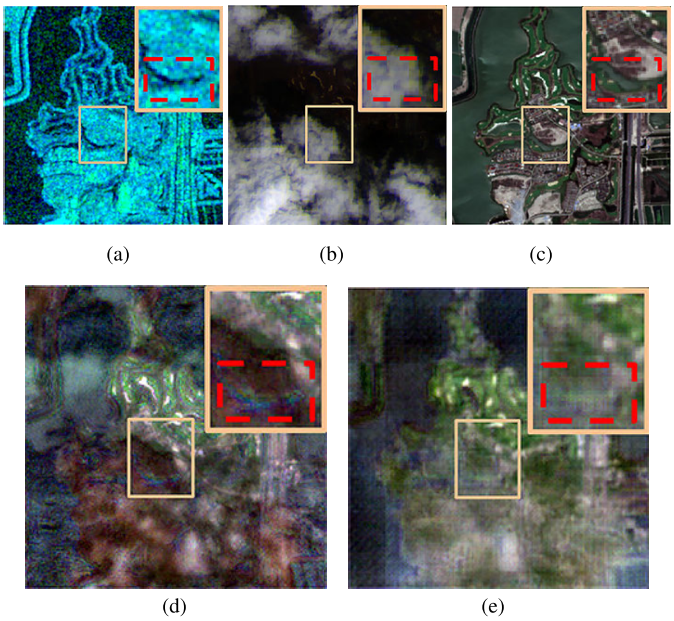


Fig. 1. Illustrative example of advanced SAR-optical data fusion cloud removal methods in situations with large cloud coverage. (From top left to bottom right) SAR image, cloudy image, cloud-free image, result of Dsen2-CR, which is the advanced CNN-based method, result of Former-CR, which is the advanced Transformer-based method. (a) SAR. (b) Cloudy. (c) Cloud-free. (d) Dsen2-CR. (e) Former-CR.

a feature extractor for both data sources simultaneously to obtain cloud-free images. However, CNNs may struggle with capturing long-range dependencies and global context within the optical data, leading to a significant visual difference between the recovery and cloud-free regions [5]. Therefore, many scholars proposed Transformer-based methods to make up for this defect. These methods incorporate the self-attention mechanism, which possesses powerful global contextual interaction capabilities, into their feature extraction stages. This integration ensures overall global consistency in both color and structure within the recovered images.

The two categories of methods have made substantial advancements in the task of SAR-optical data fusion cloud removal. However, they still face considerable challenges when dealing with images that have large cloud coverage. As shown in Fig. 1(d), the advanced CNN-based method Dsen2-CR effectively recovers the water body structure with a relatively complex contour in the red box. Nevertheless, the color of the restored green vegetation is incorrectly represented as red within the orange box. In Fig. 1(e), the advanced Transformer-based method Former-CR can maintain consistency overall in the color of the recovery images in the orange box. However, the water bodies blur boundaries with surrounding features, as shown in the red box. These indicate that both advanced methodologies failed to ensure clear contours in the cloud-removal region and consistency in color with the cloud-free region simultaneously.

Reconsidering the distinct roles of the two data sources in the cloud removal model, it becomes apparent that the model extracts local structural information about land covers beneath cloudy areas from SAR images, while obtaining spectral information about land covers in cloud-free areas from the whole optical images for recovering cloudy areas. These characteristics possibly make local features of SAR images and global features of optical images significant, which may account for the difference in cloud removal performance between CNN and Transformer-based methods. Compared to Transformers, CNNs possess superior local spatial awareness [13]. Thus, CNNs can effectively capture local structural information in SAR images, but can hardly perceive the land cover information of the optical images globally. This explains why CNN-based cloud removal methods can provide clear structures but struggle to maintain color consistency. In contrast, Transformers excel at modeling global spatial information in images and have low-pass filtering characteristics [13]. Thus, Transformers can effectively capture long-range spatial correlations of land covers in optical images, but can hardly extract detailed structural information in SAR images. This explains why Transformer-based cloud removal methods can maintain color consistency between cloud-removed and cloud-free areas, but lead to structural ambiguities in the cloud-removal images. Consequently, cloud removal methods based solely on CNNs or Transformers architectures can hardly simultaneously achieve clear land cover restoration while maintaining color consistency with cloud-free regions.

To address the issues and limitations above, a heterogeneous parallel network for cloud removal (HPN-CR) is proposed to achieve effective cloud removal in high cloud coverage scenarios. HPN-CR utilizes the proposed heterogeneous encoder with its SAR-optical input approach, where SAR data are input into the CNN branch and optical data are input into the Transformer branch. Specifically, for SAR data, HPN-CR utilizes a classic ResNet network, termed the CNN-Branch, to leverage the spatial awareness of CNN. This effectively captures local structural information, thereby enhancing the clarity of land cover contours. For optical data, HPN-CR employs the proposed Decloudformer block to construct the Decloudformer branch, termed the Transformer-Branch. This leverages the capabilities that globally model multiscale spatial correlations of the Decloudformer to ensure consistency in the spectral information. In the proposed Decloudformer block, self-attention with gated mapping (SAGM) is proposed to model the essential spatial correlations globally between cloudy and cloud-free areas. Meanwhile, the multiscale spatial attention module (MSSA) takes into account the multiscale characteristics of remote sensing objects, modeling the spatial connection between cloudy and cloud-free areas at different scales. Therefore, SAGM and MSSA collaborate to enhance the spectral consistency of similar land covers in both cloud-removed and cloud-free areas.

The contribution of this work can be summarized as follows.

1) A novel heterogeneous parallel network for SAR-optical data fusion cloud removal (HPN-CR) is proposed to achieve effective cloud removal in high cloud coverage scenarios. HPN-CR utilizes the proposed heterogeneous encoder with its SAR-optical input approach, which includes two heterogeneous branches of ResNet and the proposed Decloudformer, to extract and fuse features of SAR and optical data sources. The fused features, through a selective kernel (SK) fusion and

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pixelshuffle-based decoder, preserve cloud-free regions and achieve high-definition reconstruction from feature maps to images. Therefore, HPN-CR can achieve effective cloud removal of large-scale cloud coverage.

2) To make the detailed structure of the cloud removal recovery area clear, and the color consistent with similar land covers in the cloud-free area simultaneously, the heterogeneous encoder method with its SAR-optical input approach is proposed. The heterogeneous encoder leverages a ResNet network with local spatial awareness to extract SAR features for recovering detailed structures. It also uses the proposed Decloudformer, which globally models multiscale spatial correlations, to extract optical features for maintaining color consistency. The extracted features are then fused in the heterogeneous feature fusion block (HFFB) to realize the restoration of land cover features in cloudy areas.
3) To ensure consistency in spectral information for similar land cover types across both cloud-removed and cloud-free regions, Decloudformer is proposed to globally model multiscale spatial correlations within the input cloudy optical image. It is composed of several Decloudformer blocks, and each block contains parallel SAGM and MSSA. Specifically, SAGM introduces gated mapping into self-attention to effectively model the essential spatial correlations globally between cloudy and cloud-free areas. MSSA is proposed to model the spatial connection between cloud-covered and cloud-free areas at different scales. Therefore, SAGM and MSSA work together to provide appropriate cloud-free area land cover spectral information for cloudy areas.

The rest of this article is organized as follows. In Section II, the related works are briefly described. Section III gives a detailed description of HPN-CR. Then, Section IV validates the effectiveness of the proposed method and discusses the experimental results. Finally, the conclusion of this article is made in Section V.

II. Related Work

A. Characteristics of CNN and Transformer

The development of CNNs has greatly promoted the progress of various visual tasks, which is largely attributed to convolution operations. It extracts local features in a layered manner to form a strong image representation ability [14]. However, limited by the size of the convolution kernel and the inherent characteristics of the convolution operation, CNNs, despite their advantage in local feature extraction, struggle to capture global representations [15], for example, relationships between visual elements in long-distance contexts.

With the rise of Transformer in the natural language processing field, more and more Transformer architectures [16] are also applied to visual tasks. The vision Transformer utilizes a series of cascaded Transformer blocks to build global correlations. Leveraging both the self-attention mechanism and a multilayer perceptron (MLP) structure, the vision Transformer can portray complex spatial transformations as well as account for long-distance feature dependencies, thereby contributing to a comprehensive global representation [14]. However, the Transformer has been shown to overlook the details of local features, and researchers have suggested that this is due to its low-pass filtering characteristics [13]. Therefore, overall, CNNs are more suitable as feature extractors for SAR data sources, while Transformers are more suitable as auxiliary information extractors for optical data sources.

B. Combination of CNN and Transformer

Existing methods combining CNNs and Transformers can be coarsely categorized into two classes based on their architecture [14], [17], [18]. The first category of methods designs parallel models: a CNN branch and a Transformer branch. These methods feed the same data into both branches to obtain different feature representations. The second category of methods employs a sequential structure and replaces some modules in the Transformer with convolution-based operations. However, both approaches aim to establish strong backbone networks to enhance the representation capability for the single data source. Therefore, neither of them can be directly applied to SAR-optical data fusion cloud removal tasks. It seems intuitive to use parallel networks to receive optical and SAR imagery for cloud removal, but it is hard to determine which modality should be received by CNN and Transformer, respectively.

C. Image Restoration and Inpainting

Since image restoration and inpainting share the same goal as cloud removal, which is to improve image quality by restoring or generating missing or damaged image content, this section discusses the two related fields.

Image restoration [19] aims to recover high-quality images from their degraded counterparts (e.g., noisy, hazy, or downscaled images). Most research in image restoration focuses on developing effective analytical models and learning schemes to approximate exact mappings from degraded images to high-quality images [20]. These mapping approaches can be applied to remove thin clouds [21]. However, due to thick clouds completely obscuring the ground, these approaches are unable to establish the mapping from thick clouds to cloud-free areas, which results in an inability to effectively restore the thick cloud areas.

Image inpainting [22] aims to fill missing regions with plausible contents. Most research in image inpainting focuses on modeling the contextual information to propagate neighboring undamaged information to the holes [23]. These approaches can be applied to fill in missing information in thick cloudy areas [7]. However, they treat thin cloud regions as information-less holes, neglecting the potential information within them, which leads to an inability to effectively restore the thin cloud areas.

Therefore, the methods from the two related fields discussed above are unable to simultaneously address both thin and thick cloud regions.

III. Heterogeneous Parallel Network

HPN-CR utilizes the proposed heterogeneous encoder with its SAR-optical input approach, which includes two

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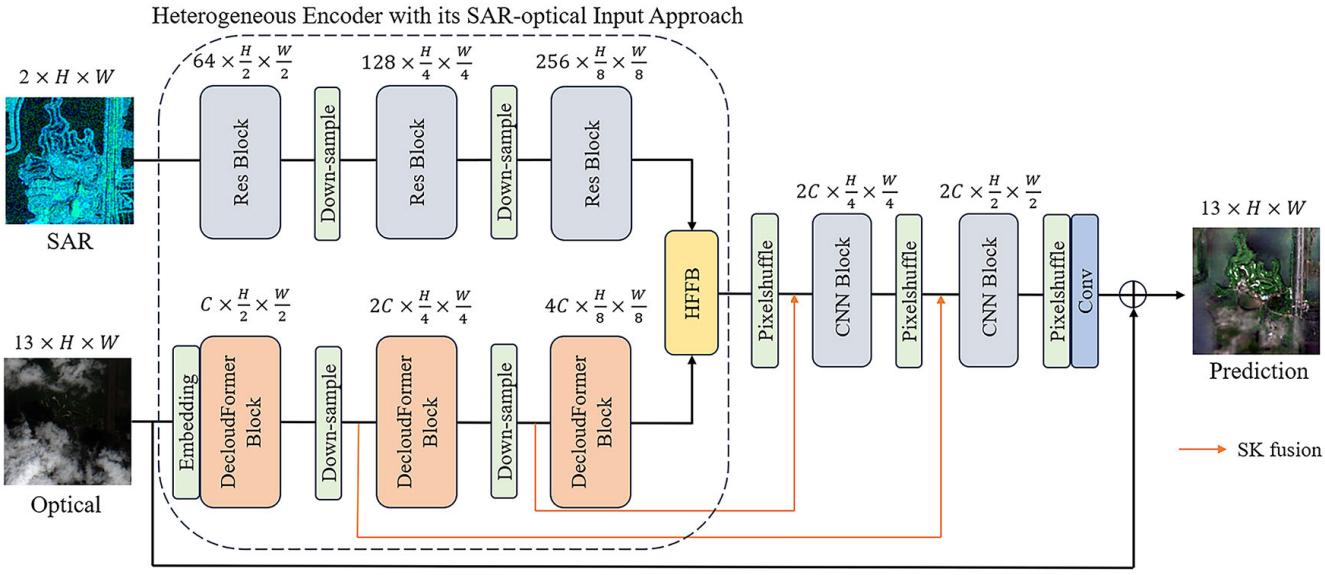


Fig. 2. Architecture of the proposed HPN-CR. The network simultaneously inputs the cloudy optical image and SAR image to the heterogeneous encoder with its SAR-optical input approach, where SAR data are input into the CNN branch and optical data are input into the Transformer branch. In the heterogeneous encoder, the CNN-Branch uses Res blocks to extract features of local structural information within the SAR image. Meanwhile, the Transformer-Branch composed of Decloudformer blocks handles the cloudy optical image, leveraging Decloudformer blocks to extract features of spectral information within optical images. The extracted features are fused in the HFFB to realize the restoration of land cover features in cloudy areas. The fused features, through an SK fusion and a pixelshuffle-based decoder, preserve cloud-free regions and achieve high-definition reconstruction from feature maps to images.

heterogeneous branches of ResNet and the proposed Decloudformer, to extract and fuse features of SAR and optical data sources. In the heterogeneous encoder, a classic ResNet network is used as CNN-Branch, taking advantage of the spatial awareness of CNN to effectively capture the local structural information from SAR images. This effectively improves the clarity of land cover contours in the cloud recovery area. For optical images, the heterogeneous encoder employs the proposed Decloudformer block to construct the Decloudformer branch as a Transformer-Branch, to globally model multiscale spatial correlations within the input cloudy optical image. This effectively ensures color consistency between cloud-removed and cloud-free areas.

A. Overview of the HPN-CR

The architecture of the proposed HPN-CR is shown in Fig. 2. HPN-CR addresses the problem of extracting diverse information from multiple heterogeneous data sources by simultaneously leveraging CNN-based and Transformer-based networks. To extract the local structural information provided in the SAR images, the CNN-Branch uses multistage Res blocks [24] that are sensitive to local spatial information [13]. In order to capture essential contextual information that can be provided by optical images, Decloudformer block is designed as an operator of Decloudformer for feature extraction from optical images. Features extracted from two branches are fused in the HFFB to get the cloud-removal features.

The input of the SK fusion and pixelshuffle-based decoder is the cloud-removal features. During the upsampling process, shallow features extracted from the optical image are concatenated to the decoder via the SK fusion layer, which is used to improve cloud-removal image quality [25]. To implement SK fusion, assume the cloud-removal features and the shallow features to be x_1 and x_2, then use a linear layer $f(\cdot)$ to project x_1 to $\hat{x}_1$. The fusion weights are formulated as follows:

$$a_1, a_2 = \text{Split}\big(\text{Softmax}\big(\mathcal{F}_{\text{MLP}}\big(\text{GAP}\big(\hat{x}_1 + x_2\big)\big)\big)\big). \quad (1)$$

Thus, the fused features of SK fusion can be represented as $y = a_1\hat{x}_1 + a_2x_2 + x_2$, where the GAP$(\cdot)$ is global average pooling, and the $\mathcal{F}_{\text{MLP}}$ is a linear–Relu–linear layer.

Pixelshuffle combined with convolution is a commonly used upsampling method to upsample feature maps, and the CNN block consists of four layers of convolution. Thus, the cloud-removal features fused with shallow features are gradually restored from feature maps to image data via the SK fusion and pixelshuffle-based decoder. To improve the quality of cloud-removal images and maximally preserve the information of cloud-free regions, the final output of the decoder is merged with the optical input image through the whole network.

B. Heterogeneous Encoder With Its SAR-Optical Input Approach

To make the detailed structure of the cloud-removal areas clear, and the color consistent with similar land covers in the cloud-free area, the heterogeneous encoder with its SAR-optical input approach is proposed. It can effectively extract and fuse the local structural information in SAR images and the spectral information in the cloud-free areas of optical images. Its SAR-optical input approach feeds the SAR data into the CNN branch and the optical data into the Transformer branch. In detail, it employs a CNN-based feature extractor to capture local structural information from SAR images. Meanwhile, it utilizes the proposed Decloudformer in parallel to capture information from optical images, which enables

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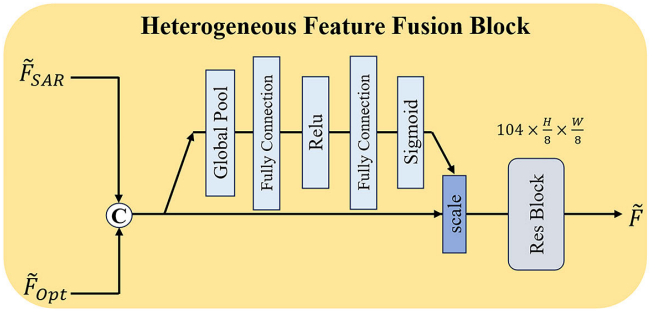


Fig. 3. Architecture of HFFB. The two inputs of this module are the extracted multispectral data source features and the extracted SAR data source features. This module first splices the two inputs in the channel dimension, and then obtains the fused features through channel attention and a Res block.

the network to learn cross modalities with different feature representations.

The CNN-Branch is designed based on the ResNet-18 [24], and the number of filters in the three stages is [64, 128, 256]. And the Transformer-Branch consists of three cascaded Decloudformer blocks. To accommodate the 13-channel multispectral data, the number of embedding layers is set to [26, 52, 104]. Through these two branches, the encoded feature maps of the two data sources can be obtained, respectively.

In the HFFB, for the purpose of enhancing the model's data fusion capability, squeeze-and-excitation (SE) [26] is employed to selectively integrate vital information from both data sources. As shown in Fig. 3, the fused feature $\hat{F}$ can be formulated as

$$\hat{F} = \mathrm{SE}\big(\mathrm{Concat}\big(\hat{F}_{\mathrm{SAR}}, \hat{F}_{\mathrm{Opt}}\big)\big). \tag{2}$$

Following SE operations, the HFFB effectively fuses features from different modalities using a Res block to realize the restoration of land cover features in cloudy areas.

C. Decloudformer

To ensure consistency in spectral information for similar land cover types across both cloudy and cloud-free regions, Decloudformer is proposed. It achieves contextual modeling of optical imagery by globally modeling multiscale spatial correlations within the input cloudy optical image. In detail, SAGM focuses on refining the self-attention mechanism's ability to model the essential spatial correlations globally between cloudy and cloud-free areas. MSSA is proposed to model the spatial connection between cloudy and cloud-free areas at different scales.

As shown in Fig. 4, the Decloudformer block performs RescaleNorm on the input optical imagery or feature map X to get X_N, replacing the traditional transformer's commonly used LayerNorm to enhance the performance of the generated images in terms of brightness and contrast [25]. Then, Decloudformer block inputs X_N into MSSA and SAGM, respectively, to get X_M

$$X_M = \mathrm{SAGM}(X_N) + \alpha \mathrm{MSSA}(X_N) + X. \tag{3}$$

In the following MLP, Gaussian error linear unit (GELU) is replaced with Leakyrelu to avoid the inverted gradient

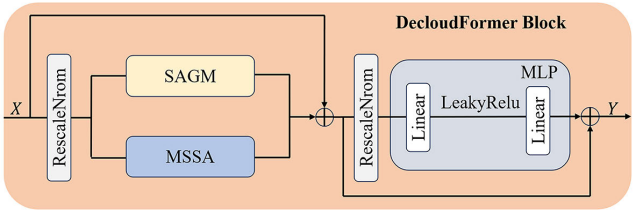


Fig. 4. Architecture of Decloudformer block. The input of this module is features of the optical data source, and the output is the features after feature extraction.

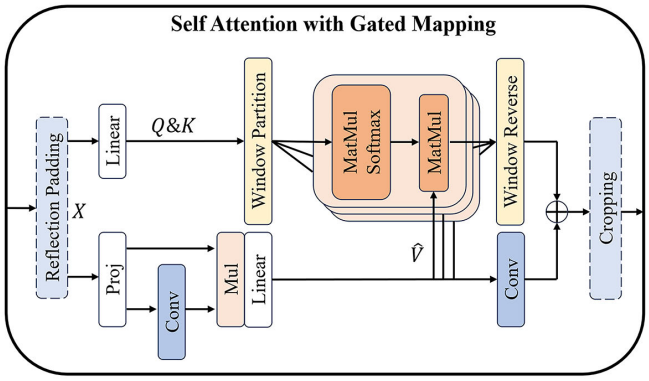


Fig. 5. Architecture of SAGM. It performs self-attention operations with gated mapping for multispectral features to model the essential spatial correlations globally between cloudy and cloud-free areas. The input of this module is features encoded from optical images, and the output is features calculated by attention operations.

introduced by GELU that affects the output of the image [25]:

$$Y = \mathrm{MLP}(\mathrm{RN}(X_M)) + X_M \tag{4}$$

where X_N and X_M denote the intermediate features. Y represents the output of the Decloudformer block.

1) Self-Attention With Gated Mapping: SAGM is proposed to model the essential spatial correlations globally between cloudy and cloud-free areas. In self-attention operations, Q and K matrices are multiplied to obtain a weight matrix, representing the correlation between different spatial locations. Subsequently, these weights are used to perform a weighted sum of the V matrix, enabling each spatial location's output to incorporate information from other locations. To help the network select essential spatial correlations, gated convolution, which is an effective way to extract essential information [27], is introduced to the V matrix as gated mapping, aiming to model the global spatial interdependencies between land cover representations in cloudy and cloud-free regions. In addition, the edge information of the image is as important as the intermediate information in the cloud removal task. Thus, reflection padding is used before the self-attention operation to make the edge pixels as important as the center pixels [25].

As shown in Fig. 5, let $X \in \mathbb{R}^{HW \times C}$ be the input feature after the reflection padding, and the output of the gated mapping $\hat{V} = \mathrm{gConv}(X)$ can be written as

$$\begin{aligned} \left[p_0^{HW \times C}, q^{HW \times C}\right] &= \mathrm{Split}\left(\phi_{\mathrm{in}}(X) \in \mathbb{R}^{HW \times C}\right) \\ p_1 &= f(q_0) \odot p_0 \in \mathbb{R}^{HW \times C} \\ \hat{V} &= \phi_{\mathrm{out}}(p1) \in \mathbb{R}^{HW \times C} \end{aligned} \tag{5}$$

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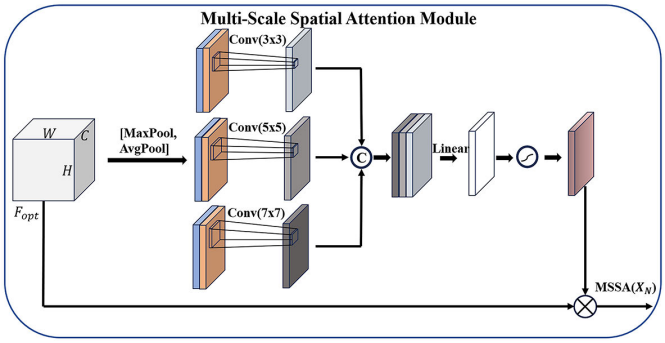


Fig. 6. Architecture of MSSA. The input of this module is features encoded from optical images, and the output is the feature representation weighted by the spatial attention weights, which can provide essential spectral information of different scale land covers in cloud-free areas for use in cloudy areas.

where ϕ_{in} and ϕ_{out} are the linear projection layers to perform channel mixing and f is a depthwise convolution. Hence, the output of the SAGM can be calculated as

$$\begin{aligned}\text{SAGM}(X_N) &= \text{Aggregation}\left(Q, K, \hat{V}\right) \\ &= \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}} + B\right)\hat{V} + \text{Conv}\left(\hat{V}\right) \end{aligned} \tag{6}$$

where $\hat{V} \in \mathbb{R}^{HW \times C}$ denotes the V before matrix multiplicating with attention map.

2) Multiscale Spatial Attention Module: MSSA is proposed to model the spatial connection between cloud-covered and cloud-free areas at different scales. In particular, multiscale spatial attention maps are extracted to enhance the perception capability of multiscale land covers.

As shown in Fig. 6, for the input features, average-pooling and max-pooling are adopted along the channel axis to generate an efficient feature descriptor, which can be expressed as

$$F_p(F) = \text{Concat}(\text{MaxPool}(F), \text{AvgPool}(F)). \tag{7}$$

On the concatenated feature descriptor F_p, three convolution layers with different kernel sizes are applied to generate multiscale spatial attention maps. Then, for the concatenated maps, a 1×1 convolution is applied to obtain enhanced spatial context information $F_s(F_p)$, which can be expressed as

$$F_s\left(F_p\right) = f^{1\times 1}\left(\text{Concat}\left(f^{3\times 3}\left(F_p\right), f^{5\times 5}\left(F_p\right), f^{7\times 7}\left(F_p\right)\right)\right). \tag{8}$$

A sigmoid function is used to get the final spatial awareness weights, and the output of the proposed MSSA can be expressed as follows:

$$\text{MSSA}(X_N) = F \otimes \text{sigmoid}\left(F_s\left(F_p\right)\right). \tag{9}$$

D. Loss Function

As shown in (10), a dedicated joint loss function is utilized to guide the training of the model. This loss function consists of three parts: SmoothL1 loss, multi-scale (MS)-structural similarity index measure (SSIM) loss [28], and spectral angle mapper (SAM) loss. SmoothL1 loss calculates the absolute value of the prediction error for each sample, then sums up all the errors, and takes the average. It can intuitively reflect the gap between the predicted value and the true value of the model, thereby making the cloud-removal image closer to the cloud-free image at the pixel level. MS-SSIM loss is an effective loss function that can be used to measure the structural similarity between images; it can improve image quality, enhance model robustness, and be consistent with human visual perception. SAM loss is based on spectral angle mapper and it is designed to guide the model to recover the correct spectral information. The joint loss function can be formulated as follows:

$$\begin{aligned}\mathcal{L}(P, T) &= \alpha L_{\text{SmoothL1}}(P, T) + (1-\alpha)L_{\text{MS-SSIM}}(P, T) \\ &\quad + \beta L_{\text{SAM}}(P, T) \end{aligned} \tag{10}$$

$$\mathcal{L}_{\text{MS-SSIM}} = 1 - [\text{MS} - \text{SSIM}(P, T)] \tag{11}$$

$$\begin{aligned}\mathcal{L}_{\text{SAM}} &= \cos^{-1} \\ &\quad \times \left(\frac{\sum_{c=h=w=1}^{C,H,W}(p)_{c,h,w} \cdot t_{c,h,w}}{\sqrt{\sum_{c=h=w=1}^{C,H,W} p_{c,h,w}^2 \cdot \sum_{c=h=w=1}^{C,H,W} t_{c,h,w}^2}}\right). \end{aligned} \tag{12}$$

Let P and T be two images with dimensions $C = 13$, $H = W = 256$, and pixel values in $[0, 1]$. The pixel values are denoted as $p_{c,h,w}$ and $t_{c,h,w}$, respectively, and use μ_p, μ_t, σ_p, σ_t, and σ_{pt} to represent their mean, standard deviation, and covariance. ϵ_1 and ϵ_2 are the small numbers for numerical stability. According to related research [28], [29], α is set to 0.2, while β is set to 0.005 to make L_{SAM} and $L_{\text{MS-SSIM}}$ comparable.

IV. Experiment and Results Analysis

A. Datasets

The SEN12MS-CR [30] was used to validate the proposed method. This dataset consists of 169 nonoverlapping regions of interest (ROIs). Each ROI covers approximately 52×40 km of ground and has a pixel size of approximately 5200×4000 pixels. To fit the network input, the ROI was cut into 256×256 pixel patches; finally, SEN12MS-CR contains 122 218 patch-triplets. Each patch-triplet is composed of cloud and cloud-free 13-band multispectral images from Sentinel-2, along with the corresponding dual-polarized (VV and VH) SAR image from Sentinel-1. The VV polarization of the SAR image was clipped to values $[-25, 0]$, while the VH polarization of the SAR image was clipped to values $[-35, 0]$. Then, they were rescaled to the range $[0, 1]$. The 13 channels of the optical image were clipped to values $[0, 10\,000]$, and then normalized to the range $[0, 1]$. To reflect the model's generalization performance, all ROIs were divided into training, validation, and test datasets in the ratio of 149:10:10 as well as [30]. The training dataset contained 107 143 patch triplets; the validation dataset contained 7176; and the test dataset contained 7899.

B. Implementation Details

1) Training Settings: The proposed HPN-CR model was implemented with the PyTorch and trained on an NVIDIA

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TABLE I
EVALUATION RESULTS ON SEN12MS-CR

Model	Input		PSNR(dB)↑	SSIM↑	SAM(°) ↓	MAE↓
	Optical	SAR				
SAR2OPT	✘	✔	23.5935	0.7828	13.8342	0.04668
GANs	✘	✔	24.1163	0.8455	9.7653	0.03792
SpA GAN	✔	✘	27.6437	0.8557	10.1822	0.03182
AMGAN-CR	✔	✘	28.1985	0.8701	9.2323	0.02991
SAR-Opt-cGAN	✔	✔	27.1266	0.8364	8.8707	0.03960
Dsen2-CR	✔	✔	28.2902	0.8623	8.9292	0.02914
GLF-CR	✔	✔	28.8497	0.8580	8.5006	0.02742
Former-CR	✔	✔	28.5932	0.8799	8.2512	0.02814
USSRN-CR	✔	✔	28.6043	0.8532	9.1736	0.02856
HSSP	✔	✔	29.0209	0.8824	8.4415	0.02718
HPN-CR(Ours)	✔	✔	**30.2273**	**0.8979**	**7.6368**	**0.02398**

A6000 graphics processing unit (GPU). The AdamW optimizer was used to train the proposed model. The initial learning rate was 0.0001, and it reduced 0.8 times every five epochs. The number of epochs and batch size were set to 15 and 32, respectively.

2) Evaluation Index: To quantitatively evaluate the performance of the proposed method, the peak-signal-to-noise ratio (PSNR), SSIM, SAM, and mean absolute error (MAE) are introduced. Their formulas are as follows:

$$\mathrm{RMSE}(x, y) = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - y_i)^2} \tag{13}$$

$$\mathrm{PSNR}(x, y) = 20 \cdot \log_{10}\left(\frac{1}{\mathrm{RMSE}}\right) \tag{14}$$

$$\mathrm{SSIM}(x, y) = \frac{(2\mu_x\mu_y + \epsilon_1)(2\sigma_{xy} + \epsilon_2)}{(\mu_x + \mu_y + \epsilon_1)(\sigma_x + \sigma_y + \epsilon_2)} \tag{15}$$

$$\mathrm{SAM}(x, y) = \cos^{-1}\frac{\sum_{i=1}^{n} x_i y_i}{\sqrt{\sum_{i=1}^{n} x_i^2 \cdot \sum_{i=1}^{n} y_i^2}} \tag{16}$$

$$\mathrm{MAE}(x, y) = \frac{\sum_{i=1}^{n}|x_i - y_i|}{n}. \tag{17}$$

The MAE is robust to outliers. Even if there are extreme values, MAE will not be distorted. The smaller the value of MAE, the more accurate the predictions. As a component of PSNR, the RMSE stands for root-mean-squared error. It is a measure of the difference between predicted values and actual values. The lower the RMSE, the better the model is. PSNR is a measure of the quality of a reconstructed image compared to the original image. The higher the PSNR, the better the quality of the reconstructed image is. SSIM is a popular image quality assessment metric that measures the perceived similarity between two images. It is designed to reflect the similarity of two images in terms of structural information, such as edges, textures, and shapes. The SAM measures the spectral fidelity of the reconstructed images.

C. Comparison With Advanced Cloud Removal Methods

This section compared HPN-CR with current advanced cloud removal algorithms: SAR2OPT [31],generative adversarial networks (GANs) [32], SpA GAN [33], and attention mechanism-based generative adversarial networks for cloud removal (AMGAN-CR) [34] are single-source cloud removal approaches; SAR-Opt-cGAN [9], Dsen2-CR [10], unified spatial–spectral sesidual network for cloud removal (USSRN-CR) [7], and hierarchical spectral and structure-preserving (HSSP) [11] are CNN-based approaches; and global–local fusion for cloud removal (GLF-CR) [5] and Former-CR [12] are Transformer-based approaches. The quantitative results are presented in Table I. The proposed HPN-CR brings remarkable improvements compared to the advanced methods. HPN-CR achieves a 1.2064-dB improvement in PSNR and also gets the best score in MAE. These two indicators can reflect that the images processed by the proposed model are closer to real images at the pixel level. For SSIM, HPN-CR achieves an improvement of 1.55%, demonstrating that the cloud-removal images achieve significant results in terms of brightness, contrast, and structural similarity. HPN-CR also achieves competitive performance on the SAM indicator, which illustrates the spectral fidelity superiority of the cloud-removal images.

The visualization results are presented in Figs. 7 and 8, and key areas are marked with red boxes. For a comprehensive and effective comparison, four different scenarios were picked out. Fig. 7 shows two sets of artificial scenes, while Fig. 8 shows two sets of nature scenes. For each scenario, from top left to bottom right are, respectively, the SAR image, the cloudy image, the cloud-free image, and the results from SAR2OPT, GANs, SpA GAN, AMGAN-CR, SAR-Opt-cGAN, Dsen2-CR, GLF-CR, Former-CR, USSRN-CR, HSSP, and HPN-CR. Comparing all the above methods horizontally, it can be seen that the proposed method has clear advantages in the visual representation of the four scenes. The detailed analysis is presented as follows.

First, four single-source cloud removal approaches were compared. SAR2OPT and GANs only apply SAR as the data source, and the images they generate can fully reflect the structural information of the ground. However, the models are unable to provide a sufficient amount of reliable spectral information. This results in a significant color discrepancy between the generated images and the actual cloud-free images, as shown in Figs. 7 and 8. SpA GAN and AMGAN-CR only use cloudy optical images as data sources to estimate potential cloud-free images. When the area of thick clouds in the image is large, the models lack sufficient information sources,

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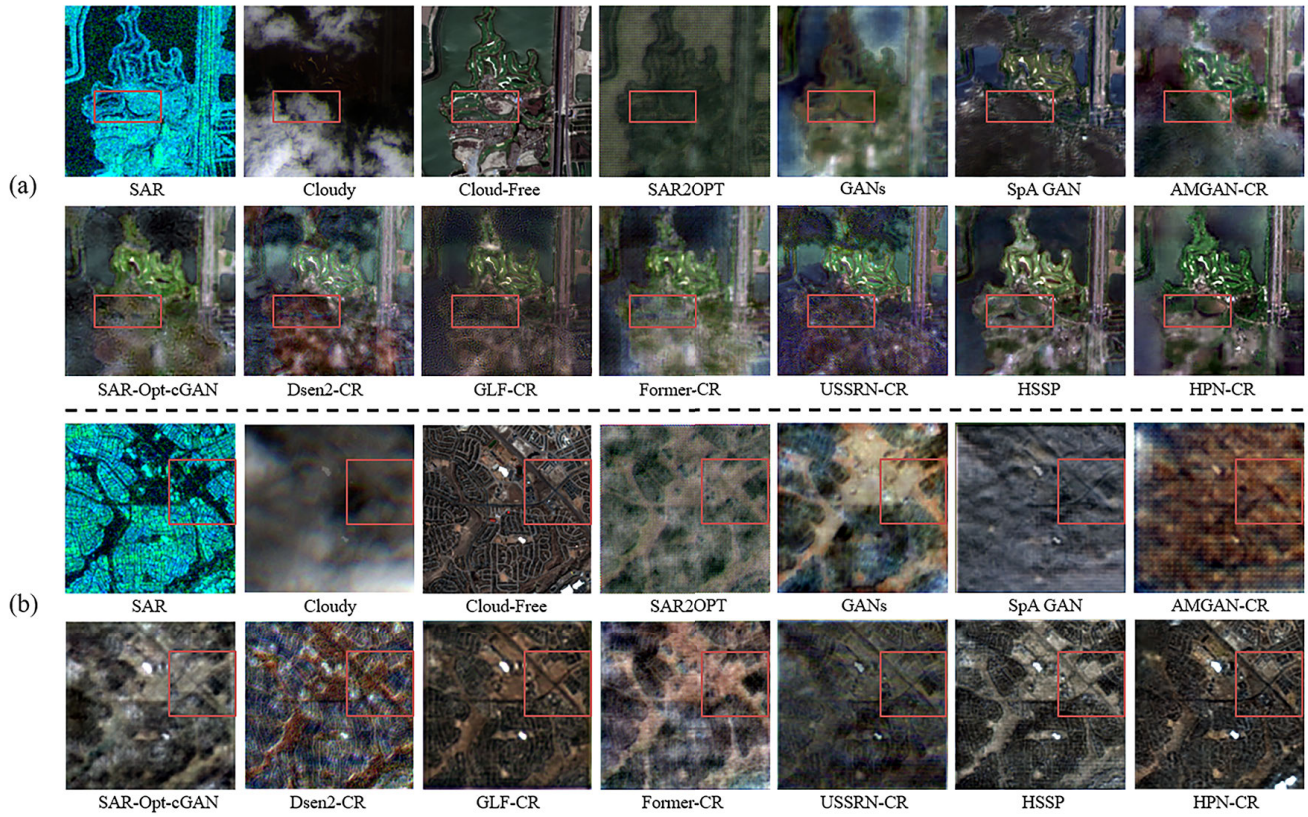


Fig. 7. Visualization results of cloud removal for two artificial scenes. (a) and (b) Represent two distinct representative instances, respectively.

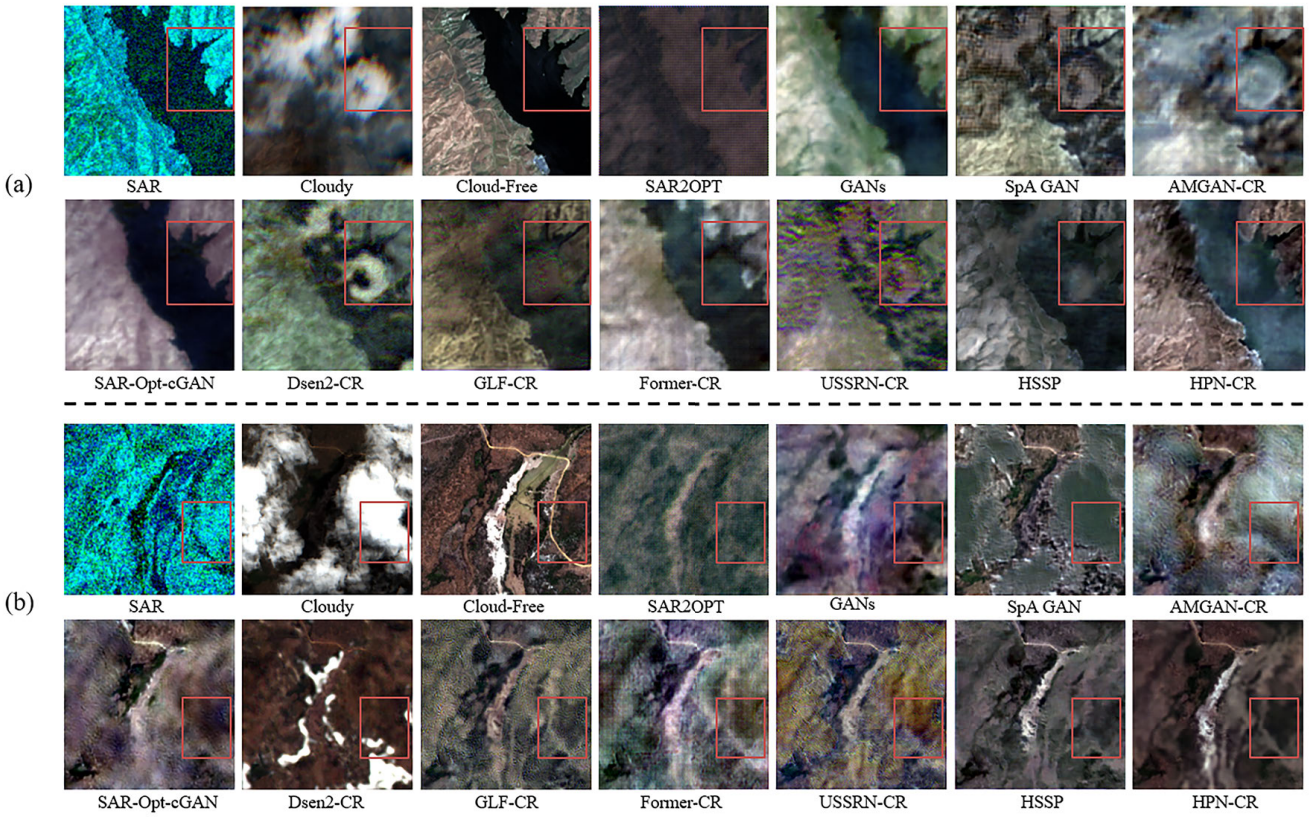


Fig. 8. Visualization results of cloud removal for two natural scenes. (a) and (b) Represent two distinct representative instances, respectively.

particularly structural information readily available from SAR images. Therefore, they can hardly estimate data obscured by cloud cover, resulting in a large area of blur in the generated image, as shown in Figs. 7(a) and 8.

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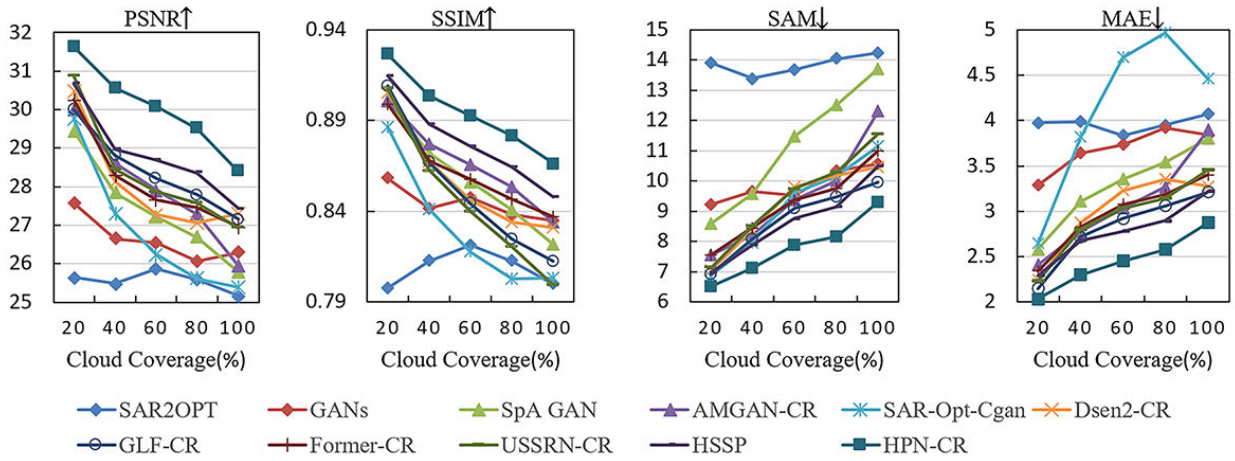


Fig. 9. Quantitative comparisons of the proposed HPN-CR to the advanced methods on different cloud cover levels.

Subsequently, four CNN-based methods were compared. They all employ SAR and optical images as data sources. As illustrated in Fig. 8(a), when the amount of clouds is large, the model almost solely relies on SAR images to recover structural information. Although the structure of the coast is restored, the overall color of the generated image is very monotonous. This means that they barely take into account the valid information in the optical input source. The image generated by the proposed method is not only structurally consistent with SAR images but also closer to real cloud-free images in color, as shown in Fig. 8. This confirms the effectiveness of Decloudformer in processing optical data sources.

Lastly, two Transformer-based methods were compared. GLF-CR and Former-CR both employ dual-source data of SAR and optical images. GLF-CR extracts the information from SAR by Transformer and exerts it to the attention map of the optical image to achieve data fusion. Former-CR directly sends the data source after channel concatenation into Uformer for end-to-end cloud removal. Since the low-pass filtering property of the Transformer makes it unable to capture the local structural information in SAR very well, as illustrated in Fig. 8(b), the local structures in the generated images are not ideally expressed. As indicated by the red boxes, Former-CR fails to recover the structures of roads and buildings, while GLF-CR, though able to restore the road structure, produces very blurry building outlines. In contrast, the proposed method effectively expresses the structures of both types of features.

Compared with all the above methods, the proposed method achieves better visual performance, and the images generated by the model are not only more realistic in structure but also recover more details and spectral information. It illustrates that the proposed method can exploit information from both data sources simultaneously to achieve cloud removal of large-scale cloud coverage.

D. Analysis of Different Cloud Cover Levels

The performance of all cloud removal methods under different cloud amounts was further analyzed by following [5]. Using the cloud detection technique of [10], the images in the test set were divided into five levels according to the cloud amount, with each 20% being level. The quantitative results can be seen in Fig. 9. The detailed analysis is as follows.

Overall, except for SAR2OPT and GANs, which use SAR as the only data source, the performance of other methods on the four indicators rapidly deteriorates as the amount of cloud cover increases, but HPN-CR can still maintain good performance. When the amount of cloud cover is less than 20%, there is little difference in the indicators of each model. This is because, in this interval, the similarity between the cloudy and cloud-free images is inherently high, so the cloud removal task only requires the model to retain the cloud-free area as much as possible. However, as the amount of cloud cover gradually increases, the advantages of the proposed method become increasingly apparent.

Specifically, since SAR2OPT only uses SAR as its data source, it is not affected by the size of the cloud cover. Moreover, when the cloud cover is greater than 60%, its performance in terms of SSIM and MAE even outperforms SAR-Opt-cGAN, once again demonstrating the important role of SAR in cloud removal tasks. Since SpA GAN only uses cloudy optical images as its data source, its performance in all metrics declines most significantly as cloud cover gradually increases. Meanwhile, it can be seen that SAR-Opt-cGAN, Dsen2-CR, GLF-CR, Former-CR, USSRN-CR, and HSSP overcome the performance degradation to some extent with the assistance of SAR. Compared with the advanced methods, the proposed method achieves the best results for each level of cloud cover.

E. Comparison With Representative CNN–Transformer Parallel Architecture Methods

This section compared HPN-CR with representative CNN–Transformer parallel architecture methods: Conformer [14] and CAF-Former [35]. To adapt to the cloud removal task, their decoders have been replaced with the decoder proposed in this article. As shown in rows 5 and 6 of Table II, the two competitive methods use the same input approach proposed in this article, where SAR data are input into the CNN branch and optical data are input into the Transformer branch. The experimental results are presented in Table II.

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TABLE II

COMPARISON OF HPN-CR WITH OTHER PARALLEL ARCHITECTURE METHODS. † DENOTES THE OPPOSITE INPUT METHOD FOR THE DATA SOURCES, WHERE SAR IS INPUT INTO THE TRANSFORMER BRANCH AND OPTICAL DATA ARE INPUT INTO THE CNN BRANCH, AS OPPOSED TO THE APPROACH PROPOSED IN THIS ARTICLE

Method	Input approch	PSNR(dB)↑	SSIM↑	SAM(°) ↓	MAE↓
Conformer†	SAR into Transformer branch Optical into CNN branch	28.2569	0.8703	9.6564	0.03021
CAF-Former†		28.2759	0.8834	8.2087	0.03048
HPN-CR†		28.8576	0.8776	8.7157	0.02928
Conformer	SAR into CNN branch Optical into Transformer branch	28.9504	0.8875	8.2865	0.02771
CAF-Former		29.1950	0.8784	8.4893	0.02742
HPN-CR(Ours)		**30.2273**	**0.8979**	**7.6368**	**0.02398**

TABLE III

QUANTITATIVE ABLATION STUDY OF THE PROPOSED HETEROGENEOUS ENCODER WITH SINGLE-BRANCH AND MULTIBRANCH NETWORKS

Model/Encoder	Data input	PSNR(dB)↑	SSIM↑	SAM(°) ↓	MAE↓
single ResNet	concatenated optical and SAR data	28.5869	0.8692	8.9438	0.02897
dual ResNet	separate optical and SAR data	29.0720	0.8838	8.0870	0.02769
single Decloudformer	concatenated optical and SAR data	29.2790	0.8937	8.0391	0.02657
dual Decloudformer	separate optical and SAR data	29.4593	0.8957	7.8337	0.02622
heterogeneous encoder (concat)	two identical concatenated optical and SAR data	30.1838	0.8959	7.6954	**0.02398**
heterogeneous encoder (proposed)	separate optical and SAR data	**30.2273**	**0.8979**	**7.6368**	**0.02398**

As shown in rows 5–7 of Table II, HPN-CR (Ours) surpasses the other competitors by at least 1.0323 dB in terms of PSNR and achieves the best score in MAE. For SSIM and SAM, HPN-CR (Ours) outperforms the other competitors by at least 1.04% and 0.6497°. These validate the superiority of HPN-CR over other CNN–Transformer architecture methods.

F. Comparison With Different Input Approaches

To validate the rationality of the proposed data source input approach, both the competitive methods and the proposed method use the opposite data source input approach for comparison, where SAR data are input into the Transformer branch and optical data into the CNN branch. They are marked † in rows 2–4 of Table II.

As shown in rows 4 and 7 of Table II, for PSNR, SSIM, and SAM, HPN-CR (Ours) outperforms HPN-CR† by 1.3697 dB, 2.03%, and 1.0789°. For the competitive methods, comparing rows 2 and 5, as well as rows 3 and 6 in Table II, the proposed input approach also demonstrates a significant performance advantage. These indicate that the proposed data source input approach is effective and reasonable.

G. Heterogeneous Encoder Analysis

To validate the effectiveness of the heterogeneous encoder with its SAR-optical input approach, where SAR data are input into the CNN branch and optical data are input into the Transformer branch, four types of encoders only based on CNN or Transformer were designed for comparison. Specifically, the heterogeneous encoder was replaced by the four encoders for cloud removal.

In this context, the CNN-based encoders consist of single and dual ResNet encoders. For the single ResNet encoder, optical and SAR data are first concatenated along the channel dimension and subsequently fed into this encoder. In contrast, for the dual ResNet encoder, optical and SAR data are separately fed into two ResNet branches. The Transformer-based encoders consist of single and dual Decloudformer encoders. For the single Decloudformer encoder, optical and SAR data are first concatenated along the channel dimension and subsequently fed into this encoder. In contrast, for the dual Decloudformer encoder, optical and SAR data are separately fed into two Decloudformer branches.

Specifically, the single and dual ResNet encoders only focus on the local features of both data sources, while the single and dual Decloudformer encoders only focus on the global features of both data sources. Therefore, none of them can combine local SAR features with global optical features. In contrast, the proposed heterogeneous encoder aims to extract and fuse local SAR features with global optical features.

As shown in rows 2–5 and 7 of Table III, the heterogeneous encoder (proposed) surpasses the other competitors by at least 0.768 dB in terms of PSNR and achieves the best score in MAE. For SSIM and SAM, the heterogeneous encoder (proposed) outperforms the other competitors by at least 0.22% and 0.1969°. These validate the effectiveness of the heterogeneous encoder. Since the proposed heterogeneous encoder aims to extract and fuse local SAR features with global optical features, it suggests that combining these features is effective.

To explore the performance of simultaneously extracting local and global features of optical and SAR, we designed a comparative method, a heterogeneous encoder (concat). The heterogeneous encoder (concat) and the heterogeneous encoder (proposed) share the same architecture. In the heterogeneous encoder (concat), optical and SAR images are first concatenated along the channel dimension. The concatenated data are then fed into the ResNet and Decloudformer branches, respectively, extracting both global and local features from

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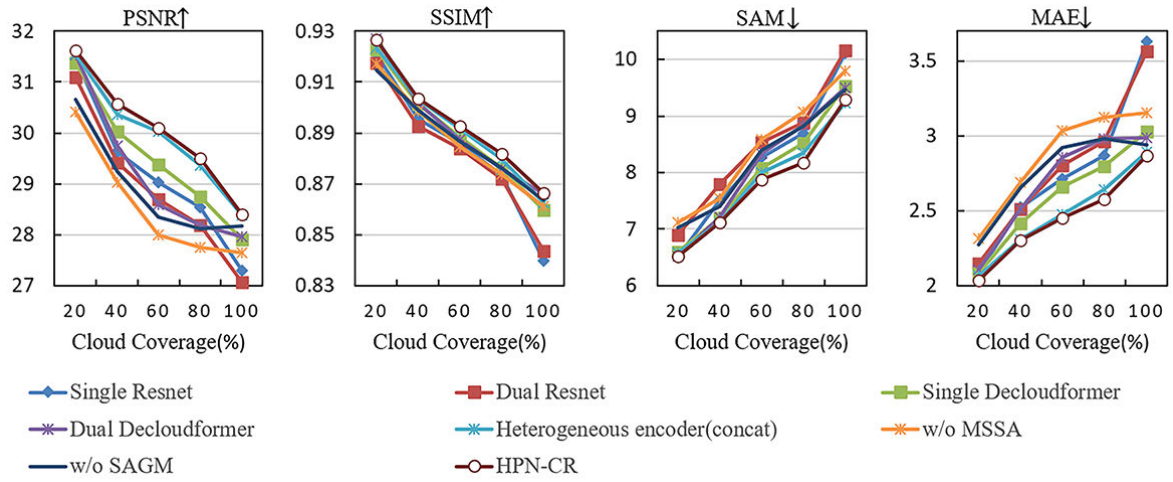


Fig. 10. Quantitative comparisons of the proposed HPN-CR to different architectures derived from HPN-CR variant and quantitative ablation study on different cloud cover levels.

TABLE IV
COMPARISON OF HFFB WITH OTHER FUSION METHODS

Method	PSNR(dB)↑	SSIM↑	SAM(°) ↓	MAE↓
addition	29.2318	0.8836	8.3982	0.0309
average	29.5791	0.8943	7.802	0.02567
concatenation	29.4748	0.8937	0.8912	0.02655
HFFB	**30.2273**	**0.8979**	**7.6368**	**0.02398**

optical and SAR data. In contrast, the proposed heterogeneous encoder aims to extract and fuse local SAR features with global optical features.

As shown in rows 6 and 7 of Table III, the heterogeneous encoder (proposed) outperforms the heterogeneous encoder (concat) by 0.0435 dB in terms of PSNR, indicating that simultaneously extracting global and local features from optical and SAR images is not well-suited for the optical-SAR fusion cloud removal task. It can be inferred that global SAR and local optical features have marginal effectiveness in cloud removal tasks.

To verify the effectiveness of HFFB, three typical fusion methods were compared, which are addition, average, and concatenation. To ensure a fair comparison, HFFB was replaced by the aforementioned fusion methods in the proposed heterogeneous encoder, respectively.

The addition fusion method performs elementwise addition of optical and SAR feature maps, the average fusion method computes the average value of these feature maps, and the concatenation fusion method concatenates these feature maps along the channel dimension. As shown in Table IV, HFFB surpasses the second competitor, average, by at least 0.6482 dB in terms of PSNR and achieves the best score in MAE. This demonstrates that HFFB's use of channel attention improves its ability to select effective features, resulting in superior performance.

H. Decloudformer Analysis

To verify the effectiveness of the proposed Decloudformer and its submodules, comparison and ablation experiments were conducted.

TABLE V
COMPARISON OF DECLOUDFORMER WITH THE ADVANCED IMAGE RESTORATION METHODS BASED ON TRANSFORMER ARCHITECTURE

Method	PSNR(dB)↑	SSIM↑	SAM(°) ↓	MAE↓
SwinIR	29.3429	0.8864	8.707	0.02734
Uformer	29.1633	0.8891	8.1814	0.02685
Restormer	29.2140	0.8749	8.5262	0.02749
Decloudformer	**30.2273**	**0.8979**	**7.6368**	**0.02398**

TABLE VI
ABLATION STUDY OF THE PROPOSED METHODS ON HPN-CR

Model		PSNR(dB)↑	SSIM↑	SAM(°) ↓	MAE↓
MSSA	SAGM				
✔	✘	29.3494	0.8921	8.0341	0.02641
✘	✔	29.0937	0.8913	8.0639	0.02741
✔	✔	**30.2273**	**0.8979**	**7.6368**	**0.02398**

1) Effectiveness of Decloudformer: Since the proposed Decloudformer can be viewed as an image restoration method, it was compared with several advanced Transformer-based image restoration methods, i.e., SwinIR [19], Uformer [36], and Restormer [37]. To ensure a fair comparison, Decloudformer was replaced by the aforementioned Transformer architectures in the proposed heterogeneous encoder, respectively. As shown in Table V, Decloudformer surpasses the other competitors by at least 0.8844 dB in terms of PSNR and achieves the best score in MAE. For SSIM and SAM, Decloudformer outperforms the other competitors by at least 0.88% and 0.5446°. These validate the effectiveness of the proposed Decloudformer.

2) Effectiveness of SAGM: To verify the effectiveness of SAGM, HPN-CR was trained by removing the SAGM in the Decloudformer block. As shown in Table VI, adopting SAGM achieves a 1.1336 dB improvement in PSNR, and SAM decreases by 0.4271°, indicating that the feature map based on gated convolution can effectively improve the performance of the self-attention mechanism, thereby improving the overall

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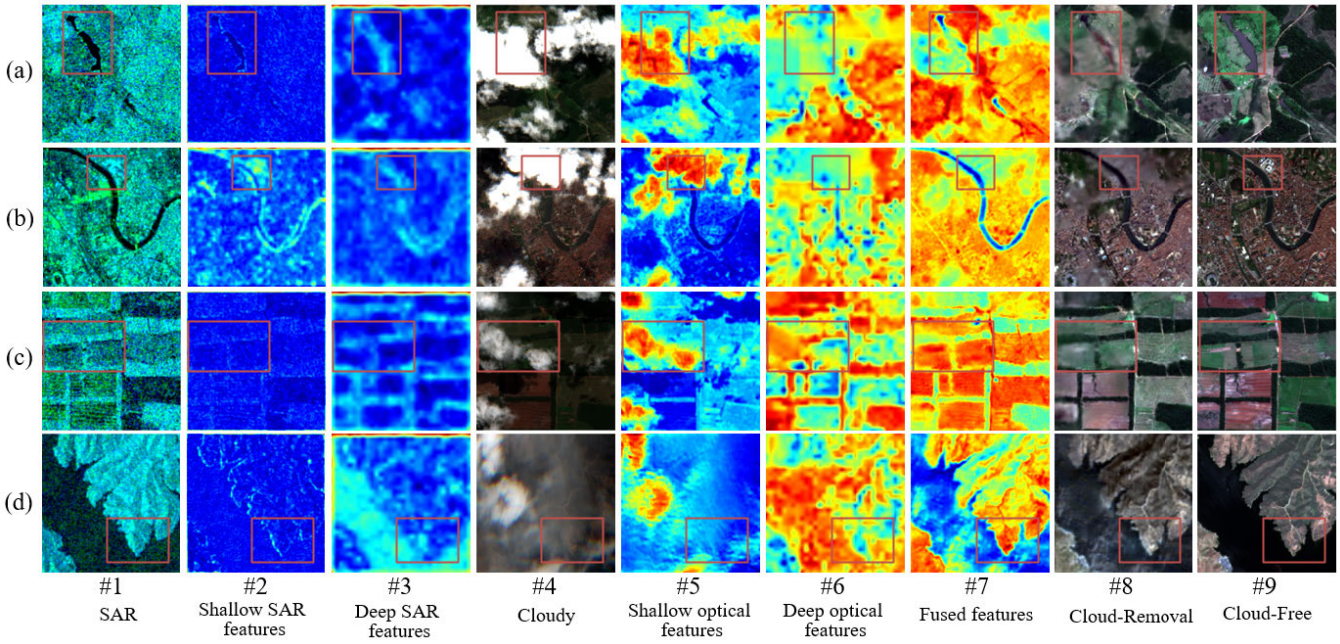


Fig. 11. Visualization results of feature maps obtained from HPN-CR. The result of the first block of each branch is taken as the shallow feature maps, and the result of the third block is taken as the deep feature maps. From left to right, is SAR images, shallow feature maps extracted from SAR, deep feature maps extracted from SAR, cloudy images, shallow feature maps extracted from optical images, deep feature maps extracted from optical images, feature maps of fused features from the result of the second CNN Block, cloud-removal images, and cloud-free images. (a)–(d) Represent four distinct representative instances, respectively.

TABLE VII
COMPARISON OF SAGM WITH OTHER GATED CONVOLUTION METHODS

Method	PSNR(dB)↑	SSIM↑	SAM(°) ↓	MAE↓
recursive gated	29.4459	0.8863	8.9241	0.02591
context gated	29.4945	0.8918	8.3249	0.02575
SAGM	**30.2273**	**0.8979**	**7.6368**	**0.02398**

TABLE VIII
COMPARISON OF MSSA WITH OTHER ATTENTION AND MULTISCALE FEATURE FUSION METHODS

Method	PSNR(dB)↑	SSIM↑	SAM(°) ↓	MAE↓
channel attention	29.5728	0.8905	8.5149	0.02547
spatial attention	29.6106	0.8871	8.7209	0.02566
ASPP	29.2628	0.8865	8.5209	0.02572
MSSA	**30.2273**	**0.8979**	**7.6368**	**0.02398**

cloud removal effect. In addition, two advanced gated convolutional methods were used for comparison, which are recursive gated convolution [27] and context gated convolution [38]. The gated mapping operation was replaced by the aforementioned gated convolutional methods in SAGM, respectively. As shown in Table VII, SAGM outperforms the other competitors by at least 0.7328 dB in terms of PSNR and achieves the best score in MAE. For SSIM and SAM, SAGM outperforms the other competitors by at least 0.61% and 0.6881°. These validate the effectiveness of the proposed SAGM.

3) Effectiveness of Multiscale Spatial Attention: To verify the effectiveness of MSSA, the HPN-CR network was trained by removing the MSSA in the Decloudformer block. As shown in Table VI, the use of MSSA improves all four indicators. As can be seen from Fig. 10, without MSSA, the performance of the model on the PSNR, which is an important metric measuring the reconstruction image quality at the pixel level, drops sharply as the cloud amount increases. However, the introduction of MSSA can largely alleviate the performance degradation caused by the increasing cloud amount. This indicates that MSSA can help the model effectively capture and utilize information from cloud-free regions, thereby improving the performance of the model when the cloud amount is high. In addition, two typical attention methods, i.e., channel attention [39] and spatial attention [39], and a multiscale feature fusion method ASPP [40], were used for comparison. To ensure a fair comparison, MSSA was replaced by the aforementioned modules in the proposed heterogeneous encoder, respectively. As shown in Table VIII, MSSA surpasses the other competitors by at least 0.6167 dB in terms of PSNR and achieves the best score in MAE. For SSIM and SAM, MSSA outperforms the other competitors by at least 0.74% and 0.8781°. These validate the effectiveness of the proposed MSSA.

I. Feature Map Analysis

To reveal the mechanism of the proposed HPN-CR method, heat maps of the optical, SAR, and fused features are visualized. Specifically, shallow and deep features of optical and SAR data in the proposed heterogenous encoder are visualized, which are extracted from the first and third blocks of each branch. The fused features output by the second CNN block in the decoder are visualized.

As shown in Fig. 11, four representative scenarios are presented. For each scenario, from left to right, each column

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represents an SAR image, shallow SAR features, deep SAR features, a cloudy image, shallow optical features, deep optical features, feature maps of fused features, a cloud-removal image, and a cloud-free image.

As illustrated in columns 1–3 of Fig. 11, both shallow and deep SAR features reveal the contours of land covers. In comparison to shallow SAR features, the contours of different land covers are more pronounced in deep SAR features. These indicate that the CNN branches effectively extracted local structural information. As illustrated in columns 4 and 5 of Fig. 11, shallow optical features highlight cloudy areas, which demonstrates that Decloudformer implicitly identified the cloudy areas. As illustrated in columns 4 and 6 of Fig. 11, by observing the deep optical features, it can be found that the dull areas are mostly thick clouds, while the bright areas are almost all clear sky or thin clouds. Since thin clouds do not completely obscure land covers, it can be inferred that Decloudformer provides spectral information of both clear sky and thin cloud regions for cloudy regions.

As illustrated in columns 4, 7, and 9 of Fig. 11, it can be found that the contours of the land-cover heatmap in the cloudy area are nearly continuous with those in the noncloud area, the heatmap brightness of the similar land covers in the cloudy and noncloud areas is also similar, and the heatmap brightness of similar land covers in both areas is also comparable. As illustrated in the eighth column of Fig. 11, the fused features in the decoder generate high-quality cloud removal results. These experimental results demonstrate that the proposed HPN-CR method can effectively combine the local structural information in SAR data and the spectral information in optical data, achieving high-performance cloud removal.

V. Conclusion

In this article, the heterogeneous parallel network for SAR-enhanced cloud removal is proposed to achieve effective cloud removal in scenarios with high cloud coverage. The proposed HPN-CR can effectively extract and integrate auxiliary information from SAR and optical data sources through a CNN and Transformer hybrid heterogeneous encoder. In the proposed heterogeneous encoder with its SAR-optical input approach, the CNN-based branch extracts local structural information from SAR, while the proposed Decloudformer-based branch extracts essential contextual spectral information from cloud-free regions. Together, they provide all the necessary auxiliary information for generating data in cloudy areas. This ensures effective cloud removal even in situations with high cloud coverage. Extensive experiments demonstrate that the proposed method achieves advanced performance in cloud removal tasks under conditions of heavy cloud cover.

However, even with SAR imagery, the structural clarity of cloud-removed optical images is sometimes insufficient. Over time, cloud-removal techniques have evolved through various stages, including interpolation, multispectral methods, and multitemporal approaches. Expanding on these developments, the optical-SAR fusion cloud removal technique adopted in this article represents a newly emerging approach in the field, with numerous aspects still warranting exploration. Notably, several well-developed remote sensing image interpretation tasks, such as object detection [41] and change detection [42], have only achieved satisfactory results after years of development.

Recently, graph neural networks (GNNs) have attracted increasing attention in low-level vision tasks [43]. Since GNNs iteratively propagate and aggregate feature information from nodes and their neighbors to learn graph representations, image restore methods based on GNNs can ideally pay attention to detail-rich regions, improving the structural clarity of restoration images. Therefore, future work will consider incorporating GNNs to increase the structural clarity of the cloud-removal images.

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